

*Lecture 4 Summary: Policy Gradient Methods

*1. Motivation

Value-based methods (e.g., Value Iteration, Q-Learning) learn value functions and then act greedily. However, they can be

The goal is to find parameters θ that maximize $J(\theta)$.

*2. Stochastic Policy Parameterizations

Setting	Example	Log-gradient
[h!] Discrete actions	Softmax over features $f_\theta(s, a)$: $\pi_\theta(a s) = \frac{e^{f_\theta(s, a)}}{\sum_{a'} e^{f_\theta(s, a')}} \nabla_\theta \log \pi_\theta(a s) = \nabla_\theta f_\theta(s, a) - \sum_{a'} \pi_\theta(a' s) \nabla_\theta f_\theta(s, a')$	
Continuous actions	Gaussian policy $\mathcal{N}(\mu_\theta(s), \sigma^2)$	$\nabla_\theta \log \pi_\theta(a s) = \frac{(a - \mu_\theta(s))}{\sigma^2} \nabla_\theta \mu_\theta(s)$

These forms make $\nabla_\theta \log \pi_\theta(a|s)$ directly computable by autodiff.

*3. Policy Gradient Theorem

The *Policy Gradient Theorem* gives a simple, model-free expression for the gradient:

Key insight: the derivative does not include $\nabla_\theta d^{\pi_\theta}(s)$, so we can estimate this gradient using on-policy samples.

*4. REINFORCE Algorithm (Williams, 1992)

A Monte Carlo estimator of the gradient:

[H] REINFORCE Initialize policy parameters θ each episode Sample trajectory $\tau = (s_0, a_0, r_0, \dots, s_T)$ under π_θ each times
REINFORCE is unbiased but has high variance.

*5. Variance Reduction via Baselines

Subtracting a baseline $b(s)$ that does not depend on a leaves the gradient unbiased:

Choosing $b(s) = V^\pi(s)$ minimizes variance and defines the *Advantage*:

The update then becomes:

*6. Actor-Critic Methods

In practice, Q^π or V^π are unknown and learned with a *critic*. The critic minimizes TD-error:

The actor then updates using δ_t as a low-variance estimate of the advantage:

*7. Average Reward (Undiscounted) Case

For continuing tasks, the objective is the average steady-state reward:

and the gradient takes the same form:

*8. Conservative and Trust-Region Policy Updates

Large policy changes can degrade performance. To guarantee monotonic improvement:

or impose a KL constraint (as in TRPO):

The practical approximation is PPO's *clipped surrogate objective*